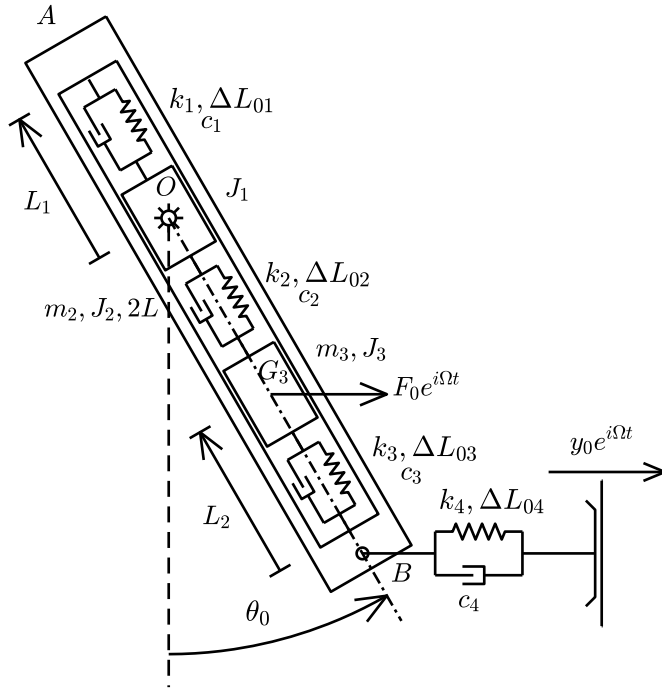


Dynamics of Mechanical Systems

November 05th, 2019



$m_2 = 5 \text{ kg}$	$k_1 = 1000 \text{ N/m}$
$m_3 = 1 \text{ kg}$	$k_2 = 1000 \text{ N/m}$
$J_1 = 0.005 \text{ kgm}^2$	$k_3 = 1000 \text{ N/m}$
$J_2 = 0.1 \text{ kgm}^2$	$k_4 = 1000 \text{ N/m}$
$J_3 = 0.005 \text{ kgm}^2$	$c_1 = 2 \text{ Ns/m}$
$L = 0.25 \text{ m}$	$c_2 = 0.5 \text{ Ns/m}$
$L_1 = 0.1 \text{ m}$	$c_3 = 0.5 \text{ Ns/m}$
$L_2 = 0.1 \text{ m}$	$c_4 = 2 \text{ Ns/m}$
$\Delta L_{01} = -0.02 \text{ m}$	$\theta_0 = \pi/6$
$\Delta L_{02} = 0.0407 \text{ m}$	
$\Delta L_{03} = 0.0322 \text{ m}$	
$\Delta L_{04} = 0.0194 \text{ m}$	

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is constituted by:

- a rectangular rigid body (J_1) connected to the ground through a hinge placed in its center of mass;
- a hollow rigid body ($m_2, J_2, 2L$) that slides with respect to the first rigid body;
- a third rectangular rigid body (m_3, J_3) that slides with respect to the hollow rigid body;
- three spring and damper systems that connect the rigid bodies and a fourth spring and damper system connected between the lower extremity of the hollow rigid body and a movable frame where a constraint displacement $y(t)$ is applied.

In the equilibrium position the angle between the vertical direction and the axis of the hollow rigid body is θ_0 .

The students are requested to:

1. Write their surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the horizontal harmonic force $F_0 e^{i\Omega t}$ at the center of mass G_3 of the third rigid body. Output: the rotation of the hollow rigid body and the relative displacement between points G_3 and B . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the vertical component of the displacement of the center of mass of the hollow rigid body and the elastic and damping force transmitted by the spring/damper system (k_4, c_4). Save the graphical results as SNxxx3.fig and SNxxx4.fig
6. Compute the time history (time step = 0.001 s) applying the harmonic constraint displacement $y_{01} e^{i(\Omega_1 t + \varphi_1)}$ and the harmonic horizontal force $F_{02} e^{i(\Omega_2 t + \varphi_2)}$ applied at point G_3 . Output: the dynamic component of the constraint force and moment transmitted between hollow rigid body and first rigid body J_1 (consider the constraint as a single slider). Save the graphical result as SNxxx5.fig and SNxxx6.fig;

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test November 05th 2019 B8.0.4* or *Test November 05th 2019 B8.0.6* or *Test November 05th 2019 B8.0.7* according to your classroom
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliare...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.